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in the thorax. The glandular cells of the midgut and hepatic or gastric caecae produce digestive juices.

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Respiratory system

The respiratory system of cockroach is well developed compared with other terrestrial insects (Figure 4.11). Branched tubes known as **trachea** open through 10 pairs of small holes called **spiracles** or **stigmata,** present on the lateral side of the body. Terminal branches of tracheal tubes are called **tracheoles** which carry oxygen to the entire body. The spiracles open and close by valves regulated by **sphincter** or **spiracular muscles**. Each tracheole is filled with a watery fluid through which exchange of gases takes place. During high muscular activity, a part of the fluid is drawn into the tissues to enable more oxygen intake and rapid diffusion. The passage of air in the tracheal system is:

SPIRACLES → TRACHEA → TRACHEOLES → TISSUES

Respiratory system of cockroach is formed of spiracles and tracheal interconnections. Why is it said to be more efficient than that of earthworm? Why inspiration of cockroach is said to be a passive process while it is an active process in man?

Circulatory system

Periplaneta has an open type of circulatory system (Figure 4.12) Blood vessels are poorly developed and opens into the haemocoel in which the blood or

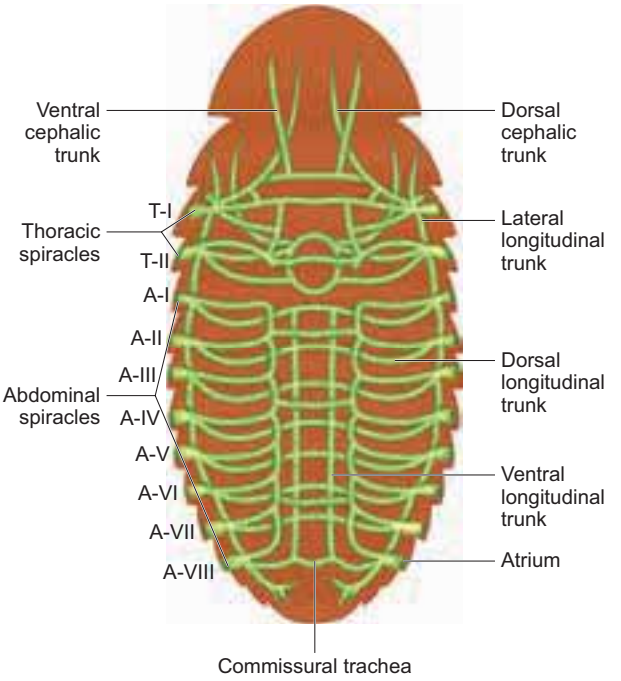


Figure 4.11 *Periplaneta americana*: Tracheal system in dorsal view

haemolymph flows freely. Visceral organs located in the haemocoel are bathed in blood. The haemolyph is colourless and consists of plasma and haemocytes which are 'phagocytic' in nature. Heart is an elongated tube with muscular wall lying mid dorsally beneath the thorax. The heart consists of 13 chambers with ostia on either side. The blood from the sinuses enters the heart through the **ostia** and is pumped anteriorly to sinuses again. The triangular muscles that are responsible for blood circulation in the cockroach are

Cockroaches survive without a head

A cockroach can live for about a week without its head. Due to their open circulatory system, and the fact that they breathe through little holes on each of their body segments, since they are not dependent on the mouth or head to breathe. The cockroach dies later due to starvation